

Bodhsanchar 1: Agent-Aided Maritime Communication System for Green Port Operations

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Abstract

This paper presents Bodhsanchar 1, an AI-enabled, agent-aided maritime communication framework designed for resilient green port operations. Addressing noise, interference, and extreme weather, the system adapts transmission power, modes, and feedback in real-time using hierarchical agents and reinforcement learning. A proposed low-cost prototype with JioFi-class devices demonstrates robust ship-to-shore links under adverse conditions. The solution enhances logistics efficiency, conserves energy, aligns with “Harit Sagar” sustainability goals, and offers scalable deployment from small vessels to large port networks.

1 Selected Innovation Theme

Smart Logistics & AI-Powered Maritime Operations.

2 Description of the Problem-Solution Fit

Maritime communications face severe challenges in dynamic ocean environments, including atmospheric interference, sea state variations, vessel motion, and extreme weather like cyclones—conditions prevalent around ports like Visakhapatnam. These disruptions lead to unreliable links, increased transmission retries, delayed vessel coordination, and higher operational risks, ultimately hindering efficient port logistics and safety.

Our solution is an AI-enabled, agent-aided communication framework that adapts feedback schemes for noisy, time-varying channels. Building on network information theory—rooted in the author’s 2006 MIT thesis on Gaussian channel reliability [1] — this system uses hierarchical agents to monitor and optimize parameters in real-time. The primary agent adjusts transmission power and modes based on channel state, while specialized agents can be included to handle weather prediction, interference management, and sea state effects using reinforcement learning.

For Phase 1 deployment, we adapt affordable JioFi-category dongles with custom software agents to create a prototype for testing adaptive transmissions. This ensures robust ship-to-shore and vessel-to-vessel links, reducing outages and enabling seamless data flow for port operations even in high sea states or storms.

3 Cost-Benefit Analysis

Implementation Costs (estimates):

- JioFi-based prototype unit: \$45 (including custom software)
- Port-side infrastructure per berth: \$1,200

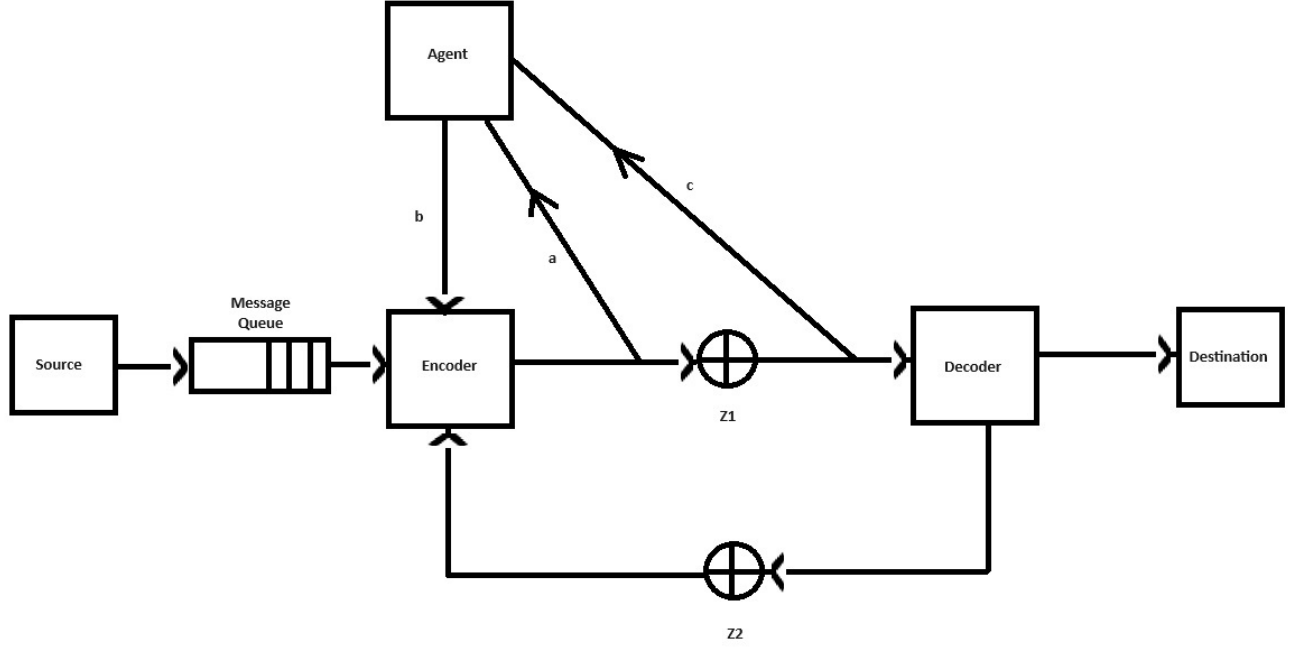


Figure 1: Agent-aided noisy feedback in Bodhsanchar 1. The agent is a simulated encoder. It determines when the encoder can act differently so as to improve performance, and correspondingly instructs the encoder. Suppose channel a can extract a fraction r of the packets from the forward channel and use these to simulate the forward link say by interpolation or regression or machine learning. In a similar way it knows the output of the forward link. Using these two and regression it monitors the communication situation. When the “ d ” of the anytime code is very large it means many many transmissions are being interleaved. In such a case the agent can instruct the encoder to repeat and complete more cycles than it is currently performing, using $\beta < d$ repetitions per cycle on average.

- Software licensing (SaaS model): \$15/vessel/month

Quantified Benefits (estimates):

- Reduced vessel waiting time: upto 25% decrease (esp. in inclement weather)
- Fuel savings from optimized scheduling: estimated lower bound \$1000/vessel/year based on waiting time reduction
- Maintenance cost reduction: minimum 30% reduction in comm-system maintenance cost due to fewer equipment failures resulting from treating delays as final failures
- Emergency response improvement: 10% faster coordination by use of AI

4 Environmental Impact Metrics (estimates)

Alignment with "Harit Sagar" (Green Ocean) initiative:

- CO₂ emission reduction estimates: 15% per port facility through optimized vessel routing using Bodhsanchar 1 comm-system
- Energy consumption decrease: 35% in communication infrastructure again due to fewer dropped calls/pings/transmissions

5 Sustainability and Scalability Potential

The framework promotes sustainability by inducing communication efficiencies that conserve power—fewer retries and optimized transmissions reduce energy consumption in vessels and port infrastructure, aligning with "Harit Sagar" goals to lower emissions and support green port initiatives. It facilitates real-time environmental monitoring (e.g., pollution tracking) and integration with renewables (e.g., reliable links for wind/solar-powered systems), minimizing fuel use through better scheduling and preventing delays that lead to idling.

Scalability is inherent in the modular design: agents can be deployed across networks, from small dongle-based setups to large-scale port systems, with extensions to MIMO or quantum protocols. We plan to commercialize via SaaS models for maritime networks, partnering with ports like VPA for pilots. This positions the solution for widespread adoption, enhancing smart logistics while driving economic and environmental impact.

References

- [1] Aman Chawla. Reliability of a gaussian channel in the presence of gaussian feedback. *S.M. Thesis*, 2006.